Technical Case Study

Rémi Patureau

1 Part A — Digital Twin (Core Physics)

1.1 Task 1 — Thermal Model

1.1.1 Statement

Build a physics-based thermal model of NovaCool. Your model should capture heat generation from servers, airflow between hot and cold aisles, and the cooling effect of the CRAH units. A lumped-parameter or reduced-order approach is preferred over CFD — we value clarity and speed over spatial resolution.

- Model each rack as a heat source whose power draw varies with a provided workload trace (CSV attached below).
- Implement a simplified airflow model: cold air enters the cold aisle at a CRAH supply temperature, absorbs heat as it passes through racks, and exits into the hot aisle.
- Model CRAH behavior: each unit removes heat from the return air using a chilled-water loop with a configurable flow rate and supply temperature.
- Simulate at 1-minute time steps over a 24-hour period.
- Produce time-series plots of: (a) average cold-aisle temperature, (b) average hot-aisle temperature, (c) peak rack outlet temperature, and (d) total cooling power consumed.

1.1.2 Solution

Different levels of approximation are possible. As a first approach, I consider that there is no thermal mass, as the inertia is very low (around 2 minutes). Two elements are modeled, the racks and the exchange with the water loop. To do so, I considered a constant efficciency ot the heat exchanger, and therefore considered the CRAH as a heat exchanger, with $\varepsilon = 0.7$.

To my understanding, the controls for the temperature are not provided, only some requirements regarding the water loop.

The elements are modeled as follows:

$$P_{rack} = \dot{Q}_{air} = \dot{m}_{air}\, Cp_{air}\, \Delta T \tag{1}$$

$$C_{min} = \min\left(\dot{m}_{air}\, Cp_{air}\ ,\ \dot{m}_{water} \times Cp_{water}\right) \tag{2}$$

$$\dot{Q}_{air} = \dot{Q}_{water} = \varepsilon\, Cmin\, (T_{in,air} - T_{in,water}) \tag{3}$$

$$T_{out,air} = T_{in,air} - \frac{\dot{Q}_{air}}{\dot{m}_c\, Cp_{air}} \tag{4}$$

$$T_{out,water} = \frac{\dot{Q}_{water}}{\dot{m}_{water}\, Cp_{water}} + T_{in,water} \tag{5}$$

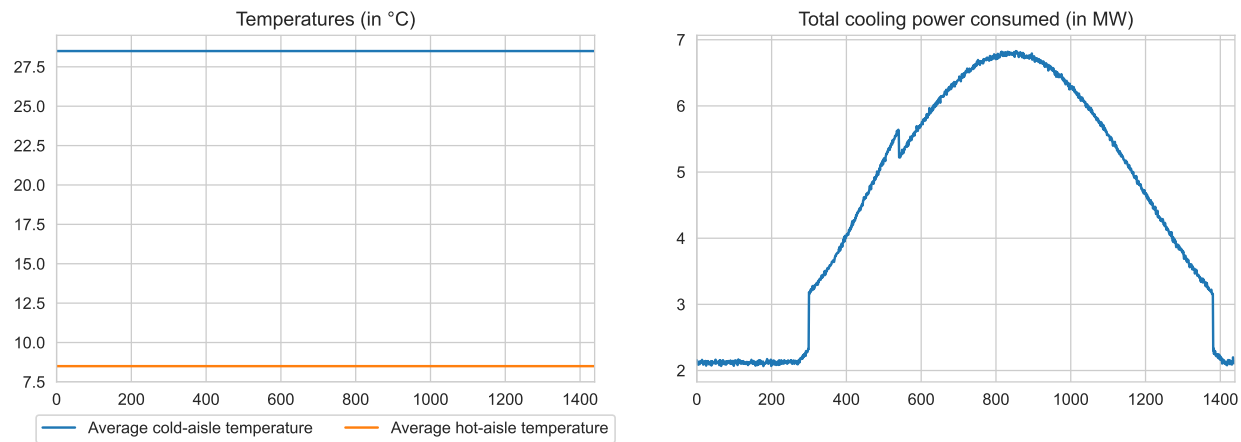

Figure 1: Test

With more time, two improvements in the modeling are possible:

•

1.2 Task 2 — Power & System Coupling

Extend your simulation to model the electrical power path and its coupling with thermal dynamics.

- Model UPS efficiency as a function of load (quadratic curve; efficiency peaks around 50–75% load).
- Include cooling infrastructure power: CRAH fans and chilled-water pumps should consume energy proportional to thermal load.
- Compute Power Usage Effectiveness (PUE) = Total Facility Power / IT Load as a time-varying metric.
- Show how PUE varies over the 24-hour period as workload and ambient conditions change.

1.3 UPS efficiency (quadratic curve)

$$\eta(x) = \eta_{peak} - k\,(x - x_{peak})^2 \tag{6}$$

With defaults $\eta_{peak} = 0.965$ (96.5% peak efficiency), $x_{peak} = 0.65$ (peak at 65% load), and k = 0.20. This means the UPS is most efficient at around 50–75% load and wastes more power at very low or very high loads.

1.4 Fan power

$$P_{fan} = P_{nominal}\, speed^3 \tag{7}$$

Pump power: Scales linearly with cooling duty (fraction of zone capacity being used), with a 15
Chiller power: Computed via a Coefficient of Performance (COP) model:

$$P_{chiller} = \frac{Q_{cooling}}{COP} \tag{8}$$

Here the COP is considered constant for this example. It should be approximated in a better way, for example via the empirical MDOE2 method.

1.5 PUE

$$PUE = (P_U PS_i nput + P_{fans} + P_{pumps} + P_{chiller} + P_{other})/P_I T \tag{9}$$

A small overhead (1% of capacity) covers lights, security, and networking.

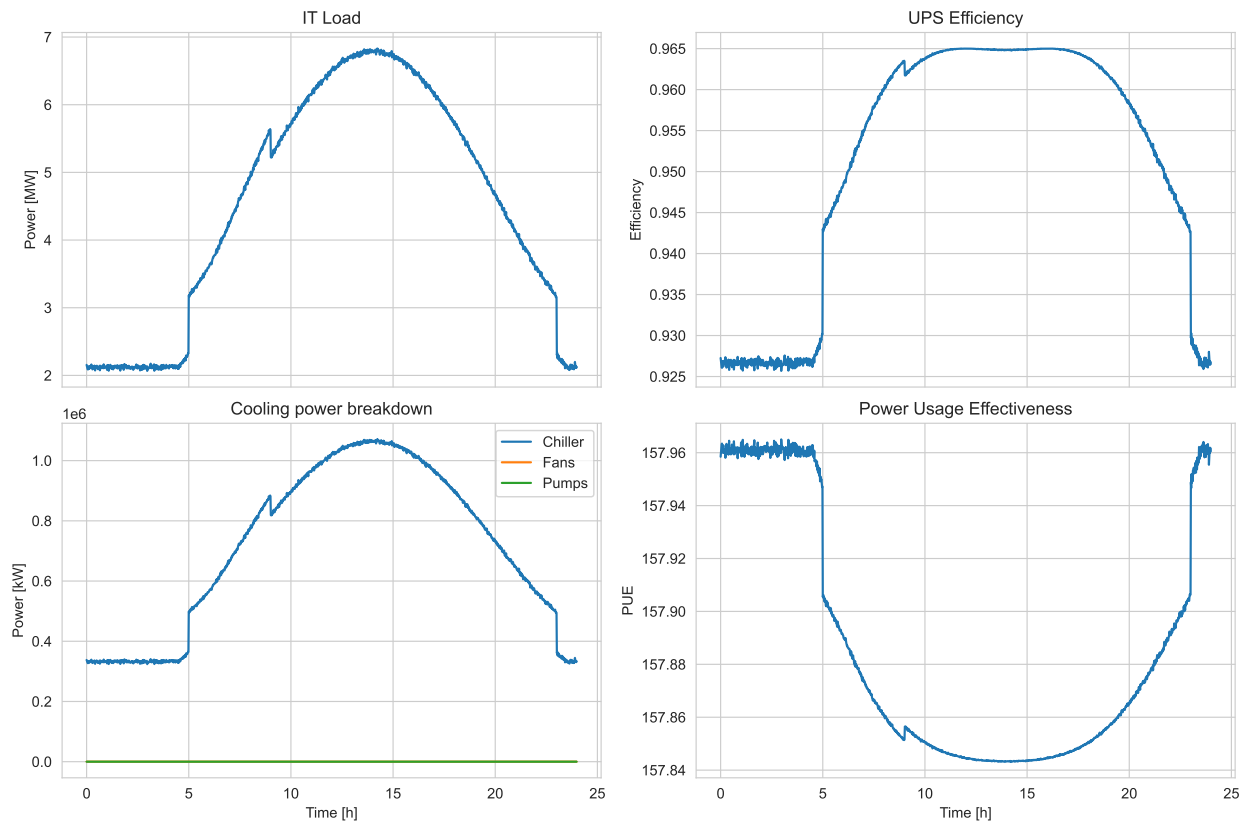

Figure 2: Thermal Model

1.6 Task 3 — Model Validation & Calibration

We provide a reference dataset (synthetic) representing 24 hours of sensor readings from NovaCool. Compare your simulation outputs against this reference.

- Compute error metrics: RMSE, MAE, and max absolute error for temperature predictions.
- Identify time intervals where your model diverges most from the reference data and hypothesize why.
- Describe at least two concrete approaches you would use to calibrate or improve model fidelity (e.g., parameter tuning, data assimilation, Bayesian calibration).

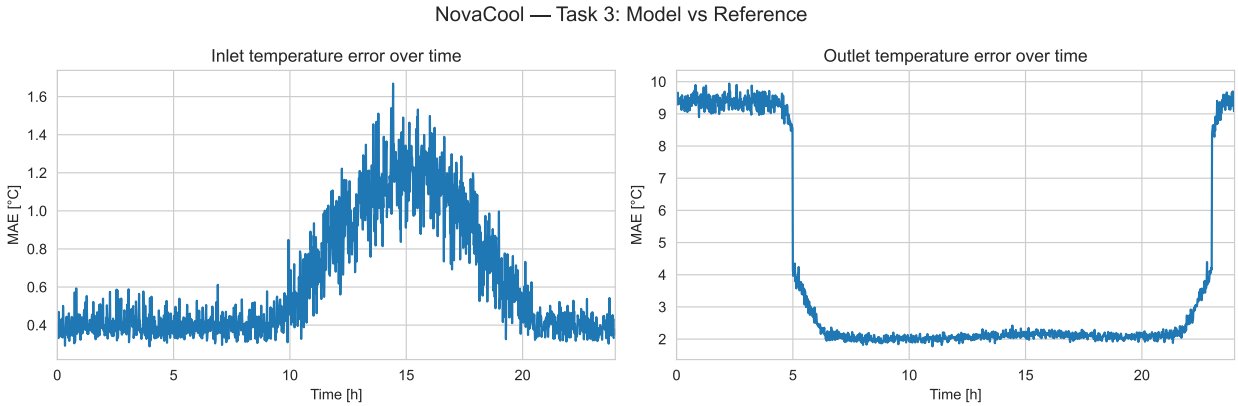

Figure 3: Temperature differences over 24h

Inlet — RMSE: 0.78 °C MAE: 0.62 °C MaxAE: 3.36 °C
Outlet — RMSE: 5.31 °C MAE: 4.00 °C MaxAE: 16.35 °C

Comparing the mass flow rate, we obtain similar results:

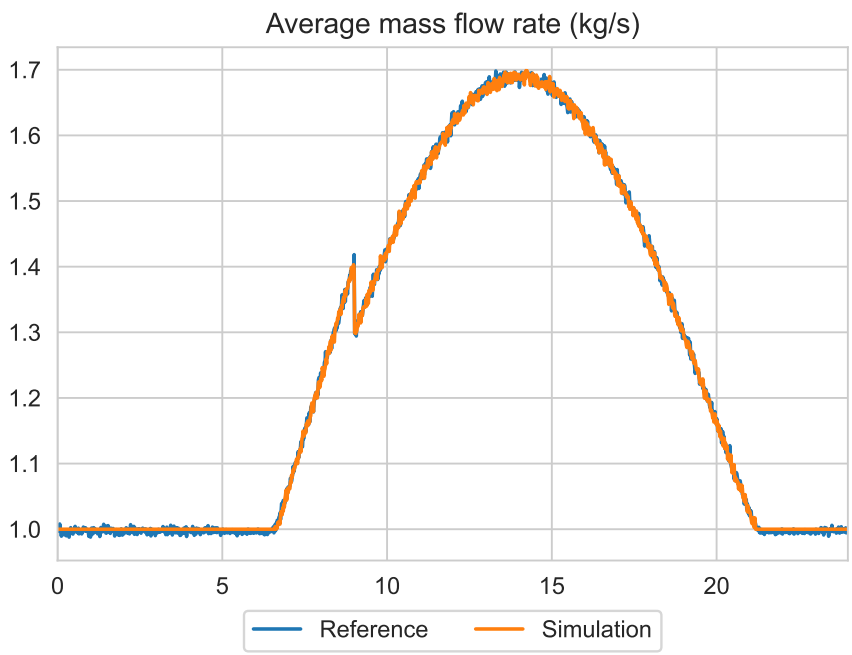

Figure 4: Mass flow rate difference over 24h

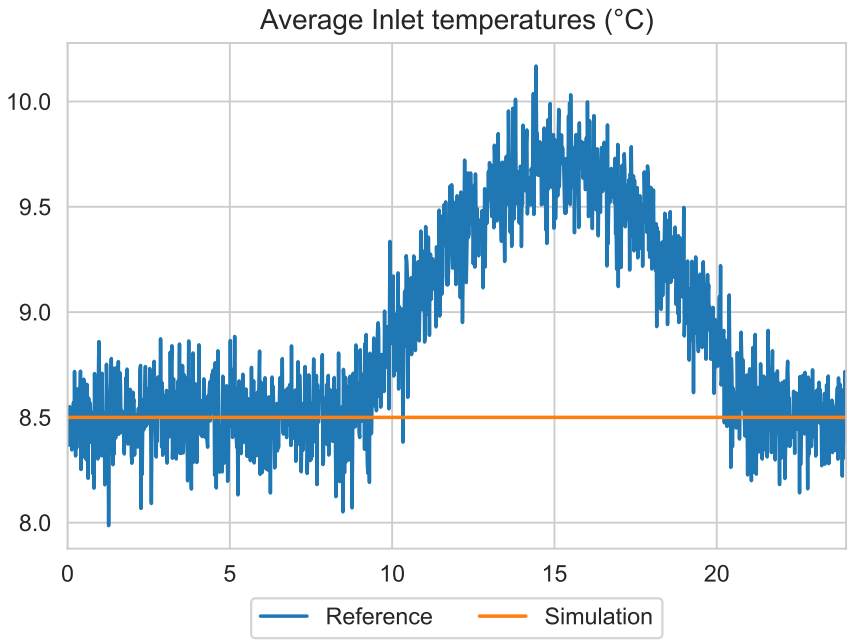

Figure 5: Inlet reference and simulated temperatures over 24h

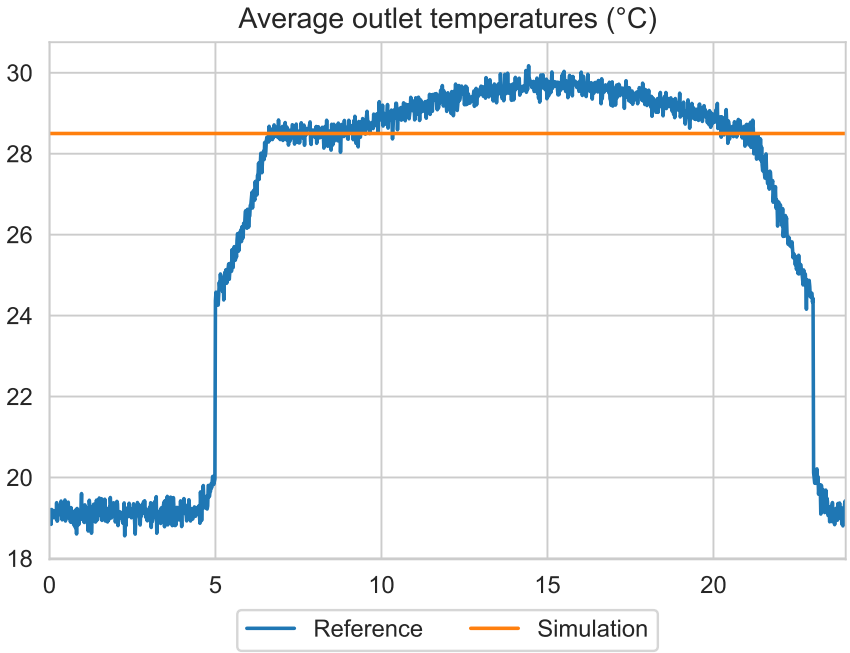

Figure 6: Outlet reference and simulated temperatures over 24h

2 Part B — Control & Environment Design

2.1 Task 4 — Control Policy

Define and implement a simple control policy for the NovaCool facility. The goal is to maximize compute throughput (i.e., allow racks to run at higher utilization) while respecting thermal safety constraints.

Objective & Constraints

- Maximize total IT load served across all racks over the 24-hour window.
- Hard constraint: No rack outlet temperature may exceed $40\,^{\circ}\mathrm{C}$ at any time step.
- Soft constraint: Minimize total cooling energy consumed (PUE should improve, not degrade).

Available Control Inputs (choose a subset)

- CRAH supply temperature setpoint (per unit)
- CRAH fan speed / airflow rate (per unit)
- Workload redistribution across racks (optional stretch goal)

Deliverables

- Describe your control policy (heuristic, PID, rule-based optimizer, or other approach).
- Run your policy on the 24-hour workload trace and compare results against a no-control baseline (fixed CRAH setpoints).
- Report: (a) total IT load served, (b) number of thermal violations, (c) total cooling energy, and (d) PUE improvement.

2.2 Task 5 — RL-Ready Environment Interface

Imagine a team of RL engineers wants to train autonomous control agents using your simulator. Wrap your digital twin as a reusable environment with a clean API. You do not need to train an RL agent — focus on the environment design.

Required Interface

Implement a minimal Gym-style interface

Define clearly:

- Observation space (st): What does the agent see? (e.g., temperatures, loads, ambient conditions, time of day)
- Action space (at): What can the agent control? (e.g., CRAH setpoints, fan speeds)
- Reward signal (rt): How is success measured? (e.g., throughput bonus minus thermal-violation penalty minus energy cost)
- Episode termination: When does an episode end? (e.g., end of 24h window, or early on critical violation)

In your write-up, justify your design choices — especially the reward function. Discuss trade-offs between reward shaping complexity and learning stability.

3 Part C — Architecture & Scaling (Discussion)

Address the following in your write-up and live presentation. No additional code is required for this section.

- **Modularity**: How do you structure the codebase so that new subsystems (e.g., a humidity model, battery storage, or a new cooling technology) can be added without rewriting existing modules?
- **Scaling**: What strategies would you use to run the simulation for a facility 10x larger (2,000 racks) and still complete a 24-hour sim run in under 5 minutes?
- **Online correction**: How would you integrate real-time sensor data into a running simulation to correct model drift (i.e., sim-to-real alignment)?
- **Testing**: Describe a testing strategy to ensure simulation correctness as the codebase evolves.
- **Sim-to-real gap**: What are the most likely sources of discrepancy between your simulation and a real facility, and how would you systematically reduce them?

3.1 Stretch Goals (Optional — for differentiation)

These are not required, but will be positively noted:

- Inject a perturbation (e.g., one CRAH fails for 30 minutes) and show how your control policy responds.
- Compare multiple control strategies (e.g., heuristic vs. PID vs. simple optimizer).
- Add stochasticity to the simulation (sensor noise, workload bursts).
- Discuss domain randomization or partial observability in the context of your environment design.